

1. What are the Products available in the market for Power BI?

Ans:

1. **Power BI Desktop** – A Windows application used to connect to data, clean and transform data, create data models, reports, and visualizations.
2. **Power BI Service** – A cloud-based platform used to publish, share, collaborate, and manage Power BI reports and dashboards.
3. **Power BI Mobile** – Mobile applications for Android and iOS that allow users to view and interact with Power BI reports and dashboards on mobile devices.
4. **Power BI Report Server** – An on-premises reporting solution that allows organizations to store and manage Power BI reports within their own infrastructure.
5. **Power BI Embedded** – A service that allows developers to integrate Power BI reports and dashboards into their own applications and websites.
6. **Power BI Report Builder** – A tool used to create paginated reports that are designed for printing and sharing in a structured format.

2. What are the different steps involved in Power BI workflow architecture?

Ans:

1. **Connect to Data** – Power BI connects to different data sources such as Excel, CSV, SQL Server, cloud databases, and online services.
2. **Data Transformation** – Power Query is used to clean and transform the data. This includes removing duplicates, handling missing values, changing data types, merging tables, and filtering data.
3. **Data Modeling** – The transformed data is organized into tables and relationships. Measures and calculated columns can also be created using DAX.
4. **Data Visualization** – Charts, tables, cards, maps, and other visualizations are created to represent the data in an understandable way.
5. **Report Creation** – Different visuals are arranged on report pages to create an interactive and meaningful report.
6. **Publish** – The completed report is published from Power BI Desktop to the Power BI Service.
7. **Share and Collaborate** – Reports and dashboards can be shared with users through workspaces, apps, or other sharing options.
8. **Monitor and Refresh** – Data can be refreshed regularly so that reports display updated information.

3. Why is Visualisation a major building block of Power BI?

Ans:

Visualization is a major building block of Power BI because it converts complex data into an easy-to-understand graphical format.

Power BI provides different visualizations such as bar charts, column charts, line charts, pie charts, maps, tables, cards, and KPI visuals.

Visualization helps users to:

- Understand large amounts of data quickly.
- Identify trends and patterns.
- Compare different categories.
- Find important business insights.
- Identify unusual values or problems.
- Make faster and better business decisions.
- Present information clearly to management and other users.

For example, a sales dashboard can use a **line chart** to show monthly sales trends and a **bar chart** to compare sales between different regions.

Therefore, visualization makes data analysis more meaningful, interactive, and easier for decision-making.

4. Explain any 3 key benefits of Power BI in the data analytics domain.

Ans:

1. Easy Data Integration

Power BI can connect to many different data sources such as Excel, CSV files, SQL databases, cloud services, and online applications. This allows analysts to bring data from different sources into one platform.

2. Interactive Data Visualization

Power BI provides many interactive charts, graphs, maps, cards, and dashboards. Users can apply filters and slicers to explore data and understand business performance easily.

3. Faster and Better Decision-Making

Power BI converts raw data into meaningful reports and dashboards. Businesses can quickly identify trends, opportunities, and problems, helping managers make data-driven decisions.

Other benefits include:

- Data cleaning and transformation using Power Query.
- Data modeling using relationships and DAX.
- Real-time and scheduled data refresh.
- Easy report sharing and collaboration.
- Accessibility through web and mobile devices.

5. What are the different responsibilities of a Power BI Developer in the market?

Ans:

A Power BI Developer is responsible for designing, developing, maintaining, and managing business intelligence reports and dashboards.

Major responsibilities include:

1. Understanding Business Requirements – Communicating with business users and understanding their reporting and analytics requirements.
2. Connecting to Data Sources – Connecting Power BI with databases, Excel files, CSV files, APIs, and other data sources.
3. Data Cleaning and Transformation – Using Power Query to clean, transform, and prepare data for analysis.
4. Data Modeling – Creating relationships between tables and designing efficient data models.
5. DAX Development – Creating calculated columns, measures, and calculations using DAX.
6. Report and Dashboard Development – Designing interactive reports and dashboards using appropriate visualizations.
7. Data Validation – Checking the accuracy and consistency of data and report results.
8. Publishing Reports – Publishing reports to Power BI Service and managing workspaces and apps.
9. Report Maintenance – Updating reports, fixing errors, and improving report performance.
10. Security Management – Implementing security features such as Row-Level Security when required.
11. Data Refresh Management – Setting up and monitoring scheduled data refreshes.
12. Performance Optimization – Improving the performance of data models, DAX calculations, and reports.

A Power BI Developer therefore works between business requirements, data, analytics, and visualization to provide useful insights to an organization.